

Anirban Bhowmick

Date of birth: 23/06/1988 | **Place of birth:** India | **Nationality:** Indian | **Phone:** (+49) 17657986552 (Mobile) | **Email address:** anirbanbhowmick88@gmail.com | **Website:** www.anirbanbhk88.github.io | **Address:** Wacholderweg, 8, 53757, Sankt Augustin, Germany (Home)

ABOUT ME

AI Engineer with 8+ years of experience bridging the gap between research (Fraunhofer IAIS) and enterprise software (TCS). Expert in RAG systems , LLM fine-tuning , and deploying scalable AI solutions.

WORK EXPERIENCE

01/12/2023 - CURRENT - SANKT AUGUSTIN, GERMANY

RESEARCH ENGINEERFRAUNHOFER IAIS

- Designed and maintained **data pipelines** for training Large Language Models (e.g., Teuken 7B). Developed data visualization tool with custom user interface, that reduced data selection time by 20%
- **Designed and scaled** an automated synthetic data generation pipeline from scratch, introducing LLM-as-a-Judge and custom metrics (faithfulness, relevance) to rigorously validate model readiness for public applications.
- **Took full ownership** of end-to-end **LLM evaluation** framework for a continuously pre-trained model optimized for German administrative data and regulated environments
- Worked on PDF file curation for document processing pipeline in a joint research project (KIMBA) with BMW to process industrial and automotive documents
- Spearheaded the extraction, preprocessing, and quality filtration **domain-specific and commercial datasets** for **RAG-based language models**
- **Co-led the implementation** of a Gaia-X node, bridging the gap between multiple enterprise partners and government stakeholders to successfully unify fragmented datasets and software resources into a centralized, federated catalog.
- Led the evaluation of enterprise data catalog tools and mentored a junior engineer to centralize fragmented training data from disjointed spreadsheets and internal pages into a unified search platform, significantly accelerating data discovery
- **Mentored students** on LLM training data curation, developing data catalogue, enhancing team productivity and knowledge sharing

GERMANY

FOUNDING ENGINEERHEALTHCARE STARTUP

- **Co-founded and acted as founding engineer** in an healthcare platform, transforming initial product concepts into a production-ready MVP by driving the development of mobile app, Python backend services, LLM API integration, DevOps pipelines
- **Led cross-functional collaboration** between medical researchers, product design, and software engineering to rapidly iterate on features based on user feedback and technical feasibility.
- **Engineered** Flutter mobile app from scratch, self hosted LLM and integration into Python backend directly translating complex product requirements

01/02/2022 - 31/10/2022 - WUPPERTAL, GERMANY

MASTER THESIS STUDENTAPTIV

- Researched and improved a LiDAR-Camera sensor fusion based deep neural network for 3D object detection in autonomous driving (ADAS)
- Automated ground truth generation thereby saving 25% of effort hours in the process

- Containerisation using **Docker and CI/CD** deployment of the trained models for use by stakeholders

01/08/2022 - 30/10/2022 - GERMANY

INTERNSHIP OMDENA

Performed **web scraping**, **data cleaning** of product data from German supermarkets and other websites, building a dashboard for displaying and comparing CO2 emissions between food categories

01/07/2021 - 31/03/2022 - HAMBURG, GERMANY

STUDENT ASSISTANT (HIWI) LANGUAGE TECHNOLOGY GROUP, UNIVERSITÄT HAMBURG

- Involved in **data preparation, Exploratory Data Analysis**, **Bi-LSTM, BERT-based Deep Learning models** for hate speech classification in movie subtitles
- Deployed the model as **Flask web app** using **Docker** and **Kubernetes** and also Open-Sourced the model the **Hugging face** library for public use
- Performed **statistical analysis** of large dataset to training Machine Learning models for learning second languages

01/09/2020 - 30/04/2021 - HAMBURG, GERMANY

WORKING STUDENT NXP SEMICONDUCTORS

- Developed Android app for IoT (“Internet of Food and Farms”)
- Peerformed critical debugging and bug fixing tasks in C++ code for IoT devices

01/12/2013 - 30/09/2019 - KOLKATA, INDIA

CONSULTANT TATA CONSULTANCY SERVICES (TCS)

- Performed predictive analysis of app usage data using deep learning **(LSTM, RNN)** algorithms and other POCs using **Statistical Machine learning (SVM, Regression)**
- Managed iOS app and SAP UI5 frontend development for KPMG's enterprise application, catering to a user base of 10,000+ individuals. Implemented Agile methodologies to drive the development process, performing code reviews, bug fixing, and elevated code quality by 50 %. Provided mentorship and guidance to junior team members
- Involved in **requirement gathering, software design, app development, technical consultation for stakeholders to deployment** of iOS app for client Phillip Morris for different markets: Ukraine, South Africa, Switzerland. Represented my team to stakeholders during on-site deputation in South Africa
- Enhanced features of hybrid mobile app to ensure compatibility across multiple devices

01/11/2011 - 30/11/2013 - KOLKATA, INDIA

SENIOR SOFTWARE DEVELOPER INDUSNET TECHNOLOGIES

- Started my career as an iOS app developer. Developed several native iOS apps
- Engaged in app design and the development of RESTful APIs for seamless app functionality. Collaborated closely with both design and backend teams to ensure effective coordination and integration

EDUCATION & TRAINING

01/10/2019 - 31/03/2023 - HAMBURG, GERMANY

MASTER OF SCIENCE - UNIVERSITÄT HAMBURG

- Machine Learning, Neural Networks, Natural language processing for web, Language technology, computer vision

Field(s) of study: Intelligent Adaptive Systems | **Final grade:** 1.9 | **Thesis:** Temporal Sensor Fusion for 3D Object Detection In Autonomous Driving | **Website:** https://www.uni-hamburg.de/

01/06/2006 - 15/07/2010 - BENGALURU, INDIA

BACHELOR OF ENGINEERING - VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Field(s) of study: Computer Science | **Website:** https://vtu.ac.in/en/

LANGUAGE SKILLS

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	C1	C1	C1	C1
GERMAN	B1	B1	B1	B1	B1

SKILLS

Python | Pytorch | SQL | Docker/CICD | shell scripting | App development | Data analysis | Data pipelines | Data preprocessing | PyTorch | SciKit-Learn | Pandas | Version Control (Git) | Code Reviews | unit testing

PUBLICATIONS

How Hateful are movies? A study and prediction of Hate content on Movies 2021

Developed *hate speech detection classifiers* using *BI-LSTM, mBERT*, and *BoW*, for classifying hate speech in movie subtitles, and established a pioneering benchmark movie dataset of 11k movie subtitles. Also, open-sourced the model in the Hugging Face library to facilitate future hate speech research within the domain of movies Developed *hate speech detection classifiers* using *BI-LSTM, mBERT*, and *BoW*, for classifying hate speech in movie subtitles, and established a pioneering benchmark movie dataset of 11k movie subtitles. Also, open-sourced the model in the Hugging Face library to facilitate future hate speech research within the domain of movies

Niklas von Boguszewski, Sana Moin, Anirban Bhowmick, Seid Muhie Yimam, Chris Biemann KONVENS Proceedings of the 17th Conference on Natural Language Processing (KONVENS 2021), Pages 37–48.

Sentiment Analysis of Low Resource Languages(Bengali) using Transformers 2021

Utilized transfer learning on Transformer architecture to develop sentiment detection models for low-resource Indian languages such as Bengali, and fine-tuned BERT classifiers on diverse datasets including wiki, news, and YouTube, subsequently validating the optimal model through A/B testing

Anirban Bhowmick, Abhik Jana ICON 2021 Proceedings of the 18th International Conference on Natural Language Processing, Pages 481–486.

PROJECTS

Eye tracking Saliency prediction system on images Built a PyTorch based Deep Learning model with a VGG16 backbone and further **fine-tuned** it with images for predicting salient regions using Transfer Learning techniques
Link https://github.com/Anirbanbhk88/Saliency-Prediction

Real Estate House Price Prediction Designed and implemented a data science project utilizing *Linear regression, decision tree regression* for *real estate price prediction*, deployed through a *Flask*-based website in Amazon EC2
Link https://github.com/Anirbanbhk88/Real-Estate-House-price-Prediction